

# Technical Appendix: MAH, Commercialization Frictions, and Original-Drug Innovation

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# 1 Appendix Roadmap and Scope

This appendix is the formal derivation layer of the MAH route-choice model. It records the complete model construction needed to support the short main text. It should be read together with the model notes, which remain the model-design control panel, and the main paper, which remains the reader-facing statement of the mechanism.

The baseline remains intentionally narrow: a partial-equilibrium dynamic route-choice model with one CMO service-market closure. The appendix does not convert the paper into a product-market GE model, an R&D input-market GE model, a complete free-entry and exit model, a welfare model, or a full bargaining model.

## 2 Model

The model combines four standard blocks rather than importing one complete canonical model. The market-return interpretation follows the pharmaceutical innovation literature in which expected market size, revenue, or profit shapes drug R&D incentives ([Acemoglu and Linn, 2004](#); [Dubois et al., 2015](#); [Barwick et al., 2026](#)). The innovation-stock and R&D-arrival structure follows product-line innovation models ([Klette and Kortum, 2004](#)). The commercialization-route choice follows the markets-for-technology and complementary-assets literature, where innovators choose between internal commercialization, cooperation, licensing, transfer, or other external routes ([Teece, 1986](#); [Arora et al., 2001](#); [Gans and Stern, 2003](#)). The property-rights interpretation follows the idea that control rights and outside options shape ex ante investment incentives ([Grossman and Hart, 1986](#); [Hart and Moore, 1990](#)). The contribution here is to adapt these blocks to the MAH setting, where the key institutional change is a regulated holder-producer separation route.

### 2.1 Scope and partial-equilibrium closure

The baseline model is a partial-equilibrium dynamic route-choice model with one market closure. It takes as given the market-return environment, the distributions of incumbent and potential entrant characteristics, the distribution of qualified producer characteristics, and the institutional regime. It endogenizes only the price of qualified contract-manufacturing services through the market-clearing condition below. Firms choose original-drug R&D intensity before successful opportunities are realized. Conditional on a successful opportunity, they choose a commercialization route.

Formally, a baseline partial-equilibrium MAH environment consists of

$$\mathcal{E}_t = \left( R_t, H(\theta_i), G_N(\theta_i), H_C(z^C), M_t, \tau^E(M_t) \right), \quad (1)$$

where  $R_t$  is the market-return shifter,  $H$  is the distribution of incumbent firm characteristics  $\theta_i$ ,  $G_N$  is the distribution of potential entrant characteristics,  $H_C$  is the distribution of qualified contract-manufacturing capacity,  $M_t$  is the institutional regime, and  $\tau^E(M_t)$  is the MAH-reducible policy and contractual friction of the entrusted route. The equilibrium service price  $p_{m,t}^*(M_t)$  is determined by

$$D_m(p_m; M_t) = S_m(p_m), \quad (2)$$

where  $D_m$  is innovator demand for qualified entrusted manufacturing services and  $S_m$  is qualified producer supply. Section 2.7 defines these objects after route choice is introduced.

The model does not solve for product-market clearing, R&D input-market clearing, free-entry zero-profit conditions, firm exit, or an invariant industry distribution. Its purpose is to characterize how MAH-induced route-set expansion, entrusted-route friction reduction, realization probabilities, and the contract-manufacturing price response affect the private opportunity value  $\Gamma_i$ , R&D intensity  $x_i^*$ , and observed realized original-drug outcomes.

Table 1: Baseline Model Closure Audit

| Object class                     | Objects                                                                                       | Status in baseline                                                                                         | Claim boundary                                                          |
|----------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Firm primitives                  | $a_i, \kappa, k_i, h_i^I, q_i^E, \mu_i^E, S_i, \tau_i^T$                                      | Taken as firm characteristics or cost parameters                                                           | Not separately identified from approval records alone                   |
| Endogenous route and R&D objects | $P_i(r   M, p_m^*), \Gamma_i(M, p_m^*), x_i^*(M), \lambda_i^{lat,*}(M), \lambda_i^{obs,*}(M)$ | Determined sequentially by route payoffs, logit route choice, CMO service-market clearing, and the R&D FOC | Observed output is downstream of latent opportunity arrival             |
| CMO market object                | $p_m^*(M)$                                                                                    | Determined by $D_m(p_m; M) = S_m(p_m)$                                                                     | Only CMO services are cleared; no product-market or R&D-input GE        |
| Composite realization objects    | $\zeta_i^E(M, p_m), \bar{R}_i^E$ , route-realization wedge                                    | Compressed objects linking latent opportunities to implementation and observed outcomes                    | Route-use and count moments discipline composites, not all primitives   |
| Calibration-facing modules       | Route use, realized original-drug counts, CMO capacity or price proxies, entry moments        | Used according to data availability                                                                        | Entry is a future or sensitivity module without a firm-year entry panel |

Table 1 is the model's closure discipline. Later results should not be read as welfare, full equilibrium, or universal innovation claims. They are claims about private route-option value and realized outcomes under the stated object status.

## 2.2 Demand side and route-independent commercialization return

The baseline model uses a route-independent gross commercialization return rather than a full product-market demand system. Let  $R_{it}^{event}$  denote the gross event payoff from

converting a successful original-drug opportunity into a marketed product before route-level implementation costs are paid. This object summarizes market size, therapeutic value, reimbursement environment, procurement exposure, and the quality or novelty composition of the opportunity. It is not the continuation value of keeping the drug in the innovating firm's future portfolio; that continuation value is represented separately by  $v_t$ .

For an entrusted-production opportunity, define the route-specific realized-return kernel

$$\bar{R}_{ip,t}^E \equiv \mathbb{E}_t[R_{ip,t}^{event} \mid L_{ip} = 1, E], \quad (3)$$

where  $L_{ip} = 1$  means that the successful opportunity is implemented and reaches production or launch through the retained entrusted route. The baseline uses  $\bar{R}_{ip,t}^E$  as a compressed payoff object. It does not separately model sales, after-sales service, pharmacovigilance, penalties, or post-launch partner-performance shocks as state variables. Appendix 8 gives a CES background interpretation for  $R^{event}$ , and Appendix 9 explains how  $\bar{R}^E$  can be viewed as the expectation of a post-production aggregate risk kernel.

The key boundary is that MAH is not modeled as shifting demand. Demand-side conditions help proxy or condition the background level of  $R^{event}$  and  $\bar{R}^E$ , but they do not identify an MAH demand effect. MAH operates through commercialization-route feasibility, route friction, realization probability, and the contract-manufacturing service price. The main MAH comparative statics hold the demand-side market-return background fixed.

## 2.3 Firm characteristics and state variables

Each firm  $i$  has fixed research and manufacturing abilities,

$$a_i > 0, \quad k_i > 0, \quad (4)$$

where  $a_i$  is research ability and  $k_i$  is manufacturing ability. The joint distribution of firm characteristics is unrestricted. Research ability and manufacturing ability may be positively correlated, negatively correlated, or independent. This is important because the model should not impose that research-oriented firms always lack manufacturing capability.

The firm also has fixed commercialization characteristics

$$(h_i^I, q_i^E, S_i, \tau_i^T, \mu_i^E).$$

The indicator  $h_i^I \in \{0, 1\}$  records whether the firm can feasibly use internal production for the opportunity. The indicator  $q_i^E \in \{0, 1\}$  summarizes whether the firm has access

to a qualified entrusted-production match when MAH makes the route legally available; in a project-producer version this object can be written as  $q_{im}^E$ . The object  $S_i - \tau_i^T$  is the net non-retained transfer outside option defined below. The object  $\mu_i^E \geq 0$  is the residual holder-side burden of using entrusted production after MAH makes the retained external route feasible: quality-control work, technical transfer, producer supervision, and cross-firm coordination that remain with the holder. This burden is not lowered mechanically by MAH in the baseline. The full incumbent distribution is therefore  $H(\theta_i)$ , where  $\theta_i = (a_i, k_i, h_i^I, q_i^E, S_i, \tau_i^T, \mu_i^E)$ .

The endogenous state is

$$n_{it} \geq 0, \quad (5)$$

where  $n_{it}$  is the stock of retained original drugs or original-drug certificates held by firm  $i$  at the beginning of period  $t$ . Existing retained drugs generate expected flow payoff

$$\pi n_{it}, \quad \pi > 0. \quad (6)$$

Here  $\pi$  should be read as  $\mathbb{E}_j[\pi_j]$ , an average retained-product flow payoff across therapeutic areas, drug ages, market sizes, and competitive environments. The stock depreciates, expires, or becomes obsolete at rate  $\delta \in (0, 1)$ . This product-stock state is a low-dimensional substitute for a full multi-product portfolio, following product-line innovation models (Klette and Kortum, 2004).

## 2.4 R&D production function and uncertainty

The firm chooses original-drug R&D intensity

$$x_{it} \geq 0. \quad (7)$$

R&D intensity produces a random number of successful original-drug opportunities in the next period. Let  $Y_{i,t+1}$  denote the number of such opportunities that reach the commercialization-route decision stage. The object  $Y_{i,t+1}$  is not a raw scientific idea arrival at the laboratory stage. In approval-side data, approvals, retained certificates, or realized original-drug records are downstream proxies for this object. A richer pipeline model could introduce an earlier idea-arrival stage and a separate clinical or regulatory success probability, but the baseline mechanism model absorbs those probabilities into effective research ability  $a_i$ . The drug-count production function is

$$\lambda_{i,t+1}(x_{it}) \equiv \mathbb{E}_t[Y_{i,t+1}] = f(a_i, x_{it}) = a_i x_{it}. \quad (8)$$

Equation (8) is the model's production function for the number of new original-drug opportunities or certificates. It is not physical pill production. It says that research

effort raises the expected number of successful opportunities, and higher research ability makes effort more productive. This is the simplest version of the stochastic innovation-arrival structure used in product-line innovation models (Klette and Kortum, 2004). A Poisson count interpretation is optional,

$$Y_{i,t+1} \mid x_{it}, a_i \sim \text{Poisson}(a_i x_{it}), \quad (9)$$

but the comparative statics below require only the conditional mean in equation (8). Any unobserved scientific success probability can be absorbed into the effective research ability  $a_i$ .

R&D effort has convex cost,

$$C(x_{it}) = \frac{\kappa}{2} x_{it}^2, \quad \kappa > 0. \quad (10)$$

This reduced-form cost delivers an interior closed-form solution. It is not a claim that project-level R&D spending is observed. It is a tractability assumption used to make the expected-return channel transparent.

## 2.5 MAH route availability and entrusted friction

Let  $M_{t+1} \in \{0, 1\}$  denote the institutional regime relevant for commercialization in period  $t + 1$ . The variable  $M_{t+1} = 0$  denotes the pre-MAH regime, and  $M_{t+1} = 1$  denotes the MAH regime. The feasible route set is firm-specific:

$$\mathcal{R}_i(M_{t+1}) = \{A, T\} \cup \{I : h_i^I = 1\} \cup \{E : M_{t+1} = 1, q_i^E = 1\}. \quad (11)$$

Here  $I$  is internal production,  $E$  is entrusted production while retaining MAH,  $T$  is a non-retained outside option such as transfer, sale, or out-licensing that removes the opportunity from the innovating firm's retained stock, and  $A$  is abandonment or delay. This discrete route set follows the commercialization-strategy logic in which innovators choose between internal development and external technology-market arrangements depending on the availability of a market for technology or ideas (Arora et al., 2001; Gans and Stern, 2003). The special case  $h_i^I = 0$  is important for research-oriented firms without internal manufacturing capacity, but it is not imposed as a universal theory assumption.

Conditional on the entrusted route being available, MAH also lowers the policy and contractual friction of using that route. A convenient way to write this is

$$\tau_{i,t+1}^E(M_{t+1}) = \bar{\tau}_i^E - \Delta\tau_i M_{t+1}, \quad \Delta\tau_i \geq 0. \quad (12)$$

Equation (12) should be read together with the route set in equation (11). When  $M_{t+1} =$

0, the MAH-style entrusted route is not feasible for the innovating holder. When  $M_{t+1} = 1$  and  $q_i^E = 1$ , the route is feasible and its friction is lower. This formulation captures both ideas: MAH adds a regulated holder-producer separation option, and the strength of the option is governed by the friction  $\tau^E$ .

The object  $\tau^E$  is deliberately route-specific and MAH-reducible. It is not the market return  $R$ , the research productivity parameter  $a_i$ , the R&D cost parameter  $\kappa$ , the direct payment for entrusted manufacturing services  $p_m$ , or the residual holder-side burden  $\mu_i^E$ . It captures institutional uncertainty, contractual responsibility allocation, and regulatory-compliance ambiguity associated with retaining authorization while using a qualified producer. A fall in  $\tau^E$  means that the external route becomes more legally supported, contractually structured, and regulatorily intelligible. It does not mean that the holder has no responsibility or that physical manufacturing, monitoring, technical transfer, and quality control become free.

For derivative comparative statics, it is useful to study  $\Gamma$  as a function of the continuous friction  $\tau^E$  conditional on  $E$  being feasible. The binary reform effect can then be decomposed into route-set expansion, a fall in  $\tau^E$ , a realization-probability response, and a contract-manufacturing price response.

## 2.6 Commercialization route choice in period $t + 1$

For each successful original-drug opportunity, the firm chooses one route  $r \in \mathcal{R}_i(M_{t+1})$ . Let  $R_{i,t+1}^{event}$  be the gross commercialization-event payoff from a commercialized original drug in period  $t + 1$ . It summarizes the non-stock payoff associated with converting a successful opportunity into a marketed product, including market size, therapeutic value, reimbursement environment, and product demand. It is included because pharmaceutical innovation responds to expected market rewards (Acemoglu and Linn, 2004; Dubois et al., 2015; Barwick et al., 2026). The model does not require firm-product revenue data.

**Payoff accounting.** There are three distinct payoff objects. First,  $R^{event}$  or  $\bar{R}^E$  is the commercialization-event payoff from turning a successful opportunity into a marketed product. Second, existing retained products generate the expected flow payoff  $\pi n_{it}$ . Third,  $v_t$  is the marginal continuation value of adding one retained original-drug asset to the firm's stock. Retained routes receive continuation value only when the opportunity is implemented and retained. Thus  $R^{event} \neq v_t$  and  $R^{event} \neq \pi n_{it}$ .

**Assumption 1** (Payoff accounting and route ownership).  *$R_{i,t+1}^{event}$  and  $\bar{R}_{ip,t+1}^E$  are route-period commercialization payoffs and exclude the continuation value of keeping the product in the innovating firm's retained stock. Continuation value is represented separately by  $v_{t+1}$ . The route  $T$  is a non-retained outside option: after transfer, sale, or non-retained out-licensing, the innovating firm receives  $S_i - \tau_i^T$  but does not add the product to its*

retained stock. A retained licensing arrangement would be a different route and is not modeled separately here.

**Non-retained transfer option.** The route  $T$  is a non-retained outside option. It includes transfer, sale, or non-retained out-licensing arrangements under which the innovating firm receives a net payment but does not add the product to its retained stock. We write this payoff as

$$b_i^T = S_i - \tau_i^T. \quad (13)$$

The object  $S_i$  is the gross expected non-retained transfer value, and  $\tau_i^T$  is the transaction or contracting friction associated with that outside option. In the baseline,  $S_i - \tau_i^T$  is taken as exogenous. This is conservative: MAH may also improve the innovator's transfer bargaining position by giving it a credible retained commercialization outside option. Appendix 12 gives a reduced-form version of this Grossman-Hart-Moore-style hold-up channel. A useful special case is

$$S_i = \omega_i^T R_{it}^{event}, \quad \omega_i^T \in [0, 1], \quad (14)$$

but the main comparative statics do not require imposing this functional form.

The entrusted route contains a composite realization probability. Let

$$\zeta_{i,t+1}^E(M_{t+1}, p_{m,t+1}) = \zeta^E(\tau_{i,t+1}^E(M_{t+1}), q_i^E, \theta_i, z^C, p_{m,t+1}) \in [0, 1], \quad (15)$$

with  $\partial \zeta^E / \partial \tau^E \leq 0$ . This object is not a separately identified primitive in the calibration. It is a compact way to distinguish latent opportunity arrival from observed realization through the retained entrusted route.

Route values in period  $t + 1$  are

$$G_{i,t+1}^I = R_{i,t+1}^{event} + v_{t+1} - C^I(k_i), \quad (16)$$

$$G_{i,t+1}^E = \zeta_{i,t+1}^E(M_{t+1}, p_{m,t+1}^*) [\bar{R}_{i,t+1}^E + v_{t+1}] - p_{m,t+1}^*(M_{t+1}) - \tau_{i,t+1}^E(M_{t+1}) - \mu_i^E, \quad (17)$$

$$G_{i,t+1}^T = S_i - \tau_i^T, \quad (18)$$

$$G_{i,t+1}^A = 0. \quad (19)$$

Here  $C^I(k_i)$  is internal production cost, with

$$\frac{\partial C^I(k_i)}{\partial k_i} < 0, \quad (20)$$

so higher manufacturing ability lowers internal production cost. Here  $p_{m,t+1}^*(M_{t+1})$  is the equilibrium project-level price or effective payment for qualified entrusted manufacturing services. It is a route-level implementation cost, not a per-unit marginal cost inside

the demand system. The separate object  $\tau_{i,t+1}^E$  captures the MAH-reducible policy and contractual friction of the retained external route. The residual burden  $\mu_i^E$  captures holder-side monitoring, quality-control, technical-transfer, and coordination costs that remain even when the route is legally supported.  $S_i - \tau_i^T$  is the net value of the non-retained outside option.

A convenient example is

$$C^I(k_i) = F_I + \frac{\bar{c}_I}{k_i}, \quad k_i > 0, \quad (21)$$

where  $F_I$  is the fixed internal commercialization burden and  $\bar{c}_I/k_i$  captures the idea that higher manufacturing capability lowers the cost of internal production. The main comparative statics require only  $C_k^I(k_i) < 0$ ; the closed-form results do not depend on this specific functional form.

Route choice is logit throughout the theory and calibration. With Type-I extreme value route shocks of scale  $\sigma_r > 0$ , the probability of route  $r$  is

$$P_i(r \mid M_{t+1}) = \frac{\exp(G_{i,t+1}^r/\sigma_r)}{\sum_{\ell \in \mathcal{R}_i(M_{t+1})} \exp(G_{i,t+1}^\ell/\sigma_r)}, \quad r \in \mathcal{R}_i(M_{t+1}), \quad (22)$$

When the CMO service price needs to be shown explicitly,  $P_i(r \mid M_{t+1}, p_{m,t+1}^*)$  denotes the same logit probability evaluated using route payoffs at the market-clearing service price. For routes other than retained entrusted production,  $\zeta_i^r(M, p_m)$  may be independent of  $p_m$ . and the expected value of one successful original-drug opportunity is the inclusive value

$$\Gamma_{i,t+1}(M_{t+1}) = \sigma_r \log \left[ \sum_{r \in \mathcal{R}_i(M_{t+1})} \exp(G_{i,t+1}^r/\sigma_r) \right]. \quad (23)$$

The Euler constant is omitted because it is an additive constant that can be absorbed into the payoff normalization. Deterministic route choice is the limiting case  $\sigma_r \rightarrow 0$ , not a separate calibration device.

## 2.7 Contract-manufacturing service market closure

The baseline closes only the market for qualified entrusted manufacturing services. Given  $p_m$  and regime  $M$ , route choice and R&D imply demand

$$D_m(p_m; M) = \int a_i x_i^*(M, p_m) P_i(E \mid M, p_m) \zeta_i^E(M, p_m) dH_i, \quad (24)$$

where  $P_i(E \mid M, p_m) = 0$  when  $E \notin \mathcal{R}_i(M)$ . Qualified producer supply is

$$S_m(p_m) = \int s_j(p_m; z_j^C) dH_C(j), \quad (25)$$

where  $z_j^C$  indexes producer-side manufacturing capability. The equilibrium service price  $p_m^*(M)$  solves equation (2). This closure lets MAH-induced entrusted-route demand raise  $p_m^*(M)$ , which can partially offset the fall in  $\tau^E$ . No other market is closed in the baseline.

## 2.8 Recursive Bellman equation

The expected retained-stock transition is

$$n_{i,t+1} = (1 - \delta)n_{it} + Y_{i,t+1} \sum_{r \in \mathcal{R}_i(M_{t+1})} P_i(r \mid M_{t+1}) \rho^r \zeta_i^r(M_{t+1}), \quad (26)$$

where  $\rho^I = \rho^E = 1$ ,  $\rho^T = \rho^A = 0$ , and  $\zeta_i^I = \zeta_i^T = \zeta_i^A = 1$  are normalizations unless richer route-specific realization data are available. Equation (26) is written in expected-count form. If one wants to track multiple realized opportunities one by one, the last term can be written as a summation over realized opportunities. That notation is not needed for the main model.

**Assumption 2** (Boundedness and stationary value). *Let  $\beta \in (0, 1)$ ,  $\delta \in (0, 1)$ ,  $\sigma_r > 0$ , and suppose route payoffs are bounded for all feasible firm types and regimes. In stationary environments, the Bellman operator is a contraction on bounded value functions, and the retained-stock component has finite marginal value because  $\beta(1 - \delta) < 1$ .*

Because the value function is affine in the stock state, write

$$V_t(n_{it}; \cdot) = A_t(\cdot) + v_t n_{it}. \quad (27)$$

The marginal value of one retained original drug satisfies

$$v_t = \pi + \beta(1 - \delta)v_{t+1}. \quad (28)$$

Using equations (8), (23), and (28), the Bellman reduces to

$$\begin{aligned} V_t(n_{it}; \cdot) = & \pi n_{it} + \beta A_{t+1} + \beta(1 - \delta)v_{t+1}n_{it} \\ & + \max_{x_{it} \geq 0} \left\{ \beta a_i x_{it} \mathbb{E}_t \left[ \Gamma_{i,t+1}(M_{t+1}, \tau_{i,t+1}^E, \mu_i^E) \right] - \frac{\kappa}{2} x_{it}^2 \right\}. \end{aligned} \quad (29)$$

The reduced problem keeps the payoff accounting transparent:  $\Gamma$  already contains the route-period payoff and the incremental continuation value of retained realized products, while  $\pi n$  is the flow payoff from the pre-existing retained stock.

### 3 Closed-Form Solution

The R&D choice solves

$$\max_{x_{it} \geq 0} \left\{ \beta a_i x_{it} \mathbb{E}_t[\Gamma_{i,t+1}] - \frac{\kappa}{2} x_{it}^2 \right\}. \quad (30)$$

The first-order condition is

$$\beta a_i \mathbb{E}_t[\Gamma_{i,t+1}] - \kappa x_{it} = 0. \quad (31)$$

Since abandonment gives value zero,  $\Gamma_{i,t+1} \geq 0$ . The optimal R&D intensity is

$$x_{it}^* = \frac{\beta a_i}{\kappa} \mathbb{E}_t[\Gamma_{i,t+1}]. \quad (32)$$

The expected number of next-period latent successful original-drug opportunities is

$$\lambda_{i,t+1}^{lat,*} \equiv \mathbb{E}_t[Y_{i,t+1}^*] = a_i x_{it}^* = \frac{\beta a_i^2}{\kappa} \mathbb{E}_t[\Gamma_{i,t+1}]. \quad (33)$$

Observed realized original-drug outcomes additionally depend on route-use probabilities and realization probabilities:

$$\lambda_{i,t+1}^{obs,*} = a_i x_{it}^* \sum_{r \in \mathcal{R}_i(M_{t+1})} P_i(r \mid M_{t+1}, p_{m,t+1}^*) \zeta_i^r(M_{t+1}, p_{m,t+1}^*). \quad (34)$$

For a research-oriented sample focused on the retained entrusted route, the corresponding object is

$$\lambda_{i,t+1}^{obs,E,*} = a_i x_{it}^* P_i(E \mid M_{t+1}, p_{m,t+1}^*) \zeta_i^E(M_{t+1}, p_{m,t+1}^*). \quad (35)$$

Equations (32)–(35) give the mechanism in closed form: expected future commercialization value raises current R&D, while observed approvals or launches are downstream realized outcomes rather than latent opportunity arrivals.

For stationary comparative statics, hold the market-return background and residual entrusted-route burden fixed, set  $M_t = M$ ,  $\tau_{it}^E = \tau_i^E(M)$ ,  $p_{m,t} = p_m^*(M)$ ,  $v_t = v$ , and  $A_t = A$ . From equation (28),

$$v = \frac{\pi}{1 - \beta(1 - \delta)}. \quad (36)$$

The stationary inclusive value under regime  $M$  is

$$\Gamma_i^M = \sigma_r \log \left[ \sum_{r \in \mathcal{R}_i(M)} \exp(G_i^r(M, p_m^*(M)) / \sigma_r) \right]. \quad (37)$$

The stationary solutions under regime  $M$  are

$$x_i^*(M) = \frac{\beta a_i}{\kappa} \Gamma_i^M, \quad (38)$$

$$\lambda_i^{lat,*}(M) = \frac{\beta a_i^2}{\kappa} \Gamma_i^M, \quad (39)$$

$$V_i(n_i; M) = \frac{\beta^2 a_i^2}{2\kappa(1-\beta)} \left[ \Gamma_i^M \right]^2 + \frac{\pi}{1-\beta(1-\delta)} n_i. \quad (40)$$

The corresponding realized-output objects are defined by equations (34) and (35).

## 4 Comparative Statics

The comparative statics hold the demand-side market-return background fixed. The main result is not that a lower generic cost raises value. The main result is a logit-weighted sorting statement: MAH creates a retained external commercialization route, and the strength of the R&D response is governed by how much probability mass the firm assigns to that route after accounting for realization probability, residual holder burdens, the endogenous contract-manufacturing price, and outside options.

### 4.1 Logit-weighted entrusted-route relevance

When  $E \in \mathcal{R}_i(M)$ , define the entrusted-route marginal payoff effect of a change in the MAH-reducible friction, holding the market-clearing service price fixed. Assume  $\bar{R}_i^E + v \geq 0$  and  $\partial \zeta_i^E / \partial \tau_i^E \leq 0$ , so a higher entrusted-route friction does not improve route realization. Then

$$\frac{\partial G_i^E}{\partial \tau_i^E} = \left( \bar{R}_i^E + v \right) \frac{\partial \zeta_i^E}{\partial \tau_i^E} - 1 \leq 0. \quad (41)$$

If  $E \notin \mathcal{R}_i(M)$ , set  $P_i(E \mid M, p_m^*(M)) = 0$ . The derivative of the inclusive value is

$$\frac{\partial \Gamma_i^M}{\partial \tau_i^E} = P_i(E \mid M, p_m^*(M)) \left[ \left( \bar{R}_i^E + v \right) \frac{\partial \zeta_i^E}{\partial \tau_i^E} - 1 \right] \leq 0. \quad (42)$$

This expression is the smooth counterpart of the deterministic sorting condition. The effect of the entrusted-route friction is small for firms that assign little route-choice probability to  $E$ , and large for firms for which  $E$  is close to or above their best alternative.

**Proposition 1** (Logit-weighted R&D response). *Conditional on the entrusted route being*

feasible, a reduction in  $\tau_i^E$  weakly raises firm  $i$ 's latent R&D opportunity arrival:

$$\frac{\partial x_i^*}{\partial \tau_i^E} = \frac{\beta a_i}{\kappa} \frac{\partial \Gamma_i^M}{\partial \tau_i^E} \leq 0, \quad (43)$$

$$\frac{\partial \lambda_i^{lat,*}}{\partial \tau_i^E} = \frac{\beta a_i^2}{\kappa} \frac{\partial \Gamma_i^M}{\partial \tau_i^E} \leq 0. \quad (44)$$

The response is stronger for high- $a_i$  firms and for firms with high  $P_i(E \mid M, p_m^*(M))$ .

*Proof.* Differentiate equations (38) and (39). The derivative of the log-sum inclusive value is the route probability times the derivative of the route payoff, which gives equation (42).  $\square$

**Proposition 2** (Entrusted-route realized-output decomposition). *For a firm with  $E \in \mathcal{R}_i(M)$ , the entrusted-route realized-output component is*

$$\lambda_i^{obs,E}(M) = a_i x_i^*(M) P_i(E \mid M, p_m^*(M)) \zeta_i^E(M, p_m^*(M)).$$

For any local change in the MAH regime or its effective intensity,

$$\begin{aligned} d\lambda_i^{obs,E} &= a_i P_i(E \mid M, p_m^*(M)) \zeta_i^E(M, p_m^*(M)) dx_i^* \\ &\quad + a_i x_i^*(M) \zeta_i^E(M, p_m^*(M)) dP_i(E \mid M, p_m^*(M)) \\ &\quad + a_i x_i^*(M) P_i(E \mid M, p_m^*(M)) d\zeta_i^E(M, p_m^*(M)). \end{aligned} \quad (45)$$

Thus observed entrusted-route outcomes can rise because MAH raises R&D intensity, shifts route probability toward  $E$ , or improves route realization. Approval-side realized outcomes should therefore not be interpreted as direct observations of latent idea arrival.

*Proof.* Differentiate equation (35).  $\square$

**Corollary 1** (Deterministic sorting as a limit). *Let  $\sigma_r \rightarrow 0$ . Define*

$$B_i^0 = \max_{r \in \mathcal{R}_i(0)} G_i^r(0), \quad E_i(1) = G_i^E(1, p_m^*(1)).$$

*Then the strict MAH response converges to the hard sorting condition*

$$\Delta_i^{MAH} > 0 \iff E_i(1) > B_i^0. \quad (46)$$

Thus the deterministic cutoff is not the baseline route-choice rule. It is the limiting interpretation of the logit model. Economically, the strongest responses should arise among high-research-capability firms with weak internal manufacturing fallback, weak non-retained transfer outside options, high entrusted-route realization probability, low residual monitoring burdens, and access to non-scarce CMO capacity.

## 4.2 Manufacturing capability and price feedback

For firms with feasible internal production, the retained external route is more attractive than internal production when

$$\zeta_i^E(M, p_m^*(M)) [\bar{R}_i^E + v] - p_m^*(M) - \tau_i^E(M) - \mu_i^E > R_i^{event} + v - C^I(k_i). \quad (47)$$

Equivalently,

$$C^I(k_i) > R_i^{event} + v - \zeta_i^E(M, p_m^*(M)) [\bar{R}_i^E + v] + p_m^*(M) + \tau_i^E(M) + \mu_i^E. \quad (48)$$

If  $\zeta_i^E = 1$  and  $\bar{R}_i^E = R_i^{event}$ , this reduces to the simpler internal-versus-entrusted cost comparison. With realization risk, the cutoff also depends on the probability that the retained entrusted route can actually be implemented.

The binary MAH effect can be discussed through the entrusted-route payoff decomposition

$$\frac{dG_i^E}{dM} = (\bar{R}_i^E + v) \frac{d\zeta_i^E}{dM} - \frac{dp_m^*(M)}{dM} - \frac{d\tau_i^E(M)}{dM} - \frac{d\mu_i^E(M)}{dM}. \quad (49)$$

This is a mechanism decomposition, not a standalone theorem. MAH raises the entrusted-route payoff when the improvement in feasibility and policy friction is large enough to dominate the contract-manufacturing price response and any residual holder-side burden. The endogenous price response is important because

$$MAH \Rightarrow D_m \uparrow \Rightarrow p_m^*(M) \uparrow$$

can partially offset the direct decline in  $\tau_i^E$ .

Table 2: Model-Implied Differences Across Mechanisms

| Mechanism                                   | Model object                                                                    | Route-use prediction                                         | Innovation prediction                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Demand expansion                            | $R^{event} \uparrow$                                                            | Does not specifically predict holder-producer split          | Multiple commercialization routes become more attractive                                                  |
| Scientific productivity improvement         | $a_i \uparrow$                                                                  | Does not specifically predict entrusted-route use            | Research-capable firms expand opportunities broadly                                                       |
| Review-speed or approval-probability reform | Lower delay or higher success probability                                       | Does not necessarily change the retained external route      | Applications or approvals can rise broadly                                                                |
| MAH route-friction channel                  | $E$ route added, $\tau_i^E \downarrow$ , $\zeta_i^E \uparrow$ , $p_m^*$ adjusts | Holder-producer split rises among relevant retained products | Strongest for high- $a_i$ , low- $k_i$ , weak transfer option, high $\zeta_i^E$ , and low $\mu_i^E$ firms |

These results are private option-value results. They do not imply that social welfare necessarily rises, that entrusted production is always chosen, or that every firm responds. The aggregate implications below are fixed-distribution sums of the firm-level logit-weighted responses, not general-equilibrium innovation or welfare statements.

## 5 Aggregate Response Modules: Incumbents, Potential Entrants, and Expected Drug Counts

Let  $H_I(\theta, n)$  be the distribution of incumbents and let  $G_N(\theta)$  be the distribution of potential entrants, where  $\theta = (a, k, h^I, q^E, S, \tau^T, \mu^E)$ . Let  $f^e$  be an entry cost with conditional cumulative distribution  $F_e(\cdot | \theta)$  and density  $f_e(\cdot | \theta)$ .

The entry block is not a full free-entry equilibrium condition in the baseline model. It is a response module that maps the post-entry value  $A(\theta, M)$  into potential research-oriented entry through an entry-cost distribution  $F_e$ . In the current approval-side data environment, this module should be interpreted as a future calibration or sensitivity exercise unless a credible firm-year entry panel is available.

The stationary entrant value, excluding old stock, is

$$A(\theta, M) = \frac{\beta^2 a^2}{2\kappa(1 - \beta)} [\Gamma^M(\theta)]^2. \quad (50)$$

A potential entrant enters if  $A(\theta, M) \geq f^e$ . Therefore the mass of entrants is

$$N_E(M) = \int F_e(A(\theta, M) | \theta) dG_N(\theta). \quad (51)$$

For a potential entrant with fixed  $\theta_i$  and entry cost  $f_i^e$ , MAH strictly changes the entry decision only when

$$\Delta \text{Entry}_i > 0 \iff f_i^e \in [A(\theta_i, 0), A(\theta_i, 1)). \quad (52)$$

Thus MAH does not make all potential entrants enter. It pushes across the entry threshold only those potential entrants whose entry costs lie between the pre-MAH and post-MAH operation values.

Conditional on MAH route availability, differentiating with respect to  $\tau^E$  gives

$$\frac{\partial A(\theta, 1)}{\partial \tau^E} = \frac{\beta^2 a^2}{\kappa(1 - \beta)} \Gamma^1(\theta) \frac{\partial \Gamma^1(\theta)}{\partial \tau^E} \leq 0. \quad (53)$$

The inequality uses  $\Gamma^1(\theta) \geq 0$ , which follows from the normalized abandonment route, and equation (42). Thus

$$\frac{\partial N_E(1)}{\partial \tau^E} = \int f_e(A(\theta, 1) | \theta) \frac{\partial A(\theta, 1)}{\partial \tau^E} dG_N(\theta) \leq 0. \quad (54)$$

This is the standard entry logic that higher post-entry value attracts additional entrants (Bresnahan and Reiss, 1991; Berry, 1992), here applied to research-oriented entry.

Holding the distribution of incumbent characteristics fixed, the model-implied latent

opportunity count from incumbents is

$$\Lambda_I^{lat}(M) = \int \frac{\beta a^2}{\kappa} \Gamma^M(\theta) dH_I(\theta, n). \quad (55)$$

The corresponding observed realized count from incumbents is

$$\Lambda_I^{obs}(M) = \int \frac{\beta a^2}{\kappa} \Gamma^M(\theta) \sum_{r \in \mathcal{R}_i(M)} P_i(r | M, p_m^*(M)) \zeta_i^r(M, p_m^*(M)) dH_I(\theta, n). \quad (56)$$

Holding the potential entrant pool fixed, the model-implied latent opportunity count from entrants is

$$\Lambda_N^{lat}(M) = \int F_e(A(\theta, M) | \theta) \frac{\beta a^2}{\kappa} \Gamma^M(\theta) dG_N(\theta). \quad (57)$$

Holding the distribution of incumbent characteristics and the potential entrant pool fixed, the model-implied aggregate latent opportunity count is

$$\Lambda^{lat,agg}(M) = \Lambda_I^{lat}(M) + \Lambda_N^{lat}(M). \quad (58)$$

This is an aggregate response under fixed distributions, not a general-equilibrium aggregate obtained from an invariant industry distribution. Conditional on MAH route availability, the incumbent-margin derivative is

$$\frac{\partial \Lambda_I^{lat}(1)}{\partial \tau^E} = \int \frac{\beta a^2}{\kappa} \frac{\partial \Gamma^1(\theta)}{\partial \tau^E} dH_I(\theta, n) \leq 0. \quad (59)$$

The entrant-margin derivative is

$$\begin{aligned} \frac{\partial \Lambda_N^{lat}(1)}{\partial \tau^E} &= \int F_e(A(\theta, 1) | \theta) \frac{\beta a^2}{\kappa} \frac{\partial \Gamma^1(\theta)}{\partial \tau^E} dG_N(\theta) \\ &+ \int \frac{\beta a^2}{\kappa} \Gamma^1(\theta) f_e(A(\theta, 1) | \theta) \frac{\partial A(\theta, 1)}{\partial \tau^E} dG_N(\theta) \leq 0. \end{aligned} \quad (60)$$

If the optional entry module is activated, both entrant-margin terms are weakly negative by equation (42) and equation (53). Therefore, within that extended module,

$$\frac{\partial \Lambda^{lat,agg}(1)}{\partial \tau^E} \leq 0. \quad (61)$$

A reduction in entrusted-production friction weakly increases the fixed-distribution aggregate response through incumbent intensity, and through potential entrant response only when the entry module is explicitly used.

**Corollary 2** (Baseline fixed-incumbent implication). *For any fixed distribution of incumbent firm characteristics, a net increase in  $\Gamma^M(\theta)$  weakly raises model-implied aggregate expected original-drug opportunities from incumbent firms. Conditional on the entrusted*

route being feasible and holding the CMO price feedback fixed,

$$\frac{\partial \Lambda_I^{lat}(1)}{\partial \tau^E} \leq 0, \quad (62)$$

so  $d\tau^E < 0$  implies  $d\Lambda_I^{lat} \geq 0$  through the friction channel. With endogenous CMO pricing, the same conclusion requires the price response in  $p_m^*(M)$  not to offset the entrusted-route value gain. The increase is strict if a positive-measure set of incumbents has  $a > 0$ , positive inclusive-value response, and positive entrusted-route probability. If a credible firm-year entry panel is available and the optional entry module is activated, the same sign logic can be extended to potential entrants through  $F_e$ , but that is not a baseline empirical conclusion.

*Proof.* Equation (59) implies equation (62). Strictness follows when  $\partial \Gamma^1 / \partial \tau^E < 0$  on a positive-measure set of incumbent firms. Equation (60) gives the analogous optional entry-module term.  $\square$

## 6 Mechanism Calibration Blueprint

The model is not intended to identify every primitive. The calibration exercise is a mechanism calibration: it asks whether data moments close to the route-choice and realization channels can discipline a small number of composite objects. It is not a full structural estimation exercise and not a stand-alone causal identification design.

### 6.1 Model-to-data mapping

The key requirement is to keep model objects, empirical moments, and claim strength separate. Table 3 summarizes the mapping.

### 6.2 Executable calibration steps

A feasible blueprint has five steps.

**Step 1: Define the sample and moments.** Construct a comparable product-level sample of original-drug records. If available, also construct demand-side controls  $d_t^{data}$  for  $R^{event}$  and  $\bar{R}^E$ , such as disease burden, patient population, reimbursement status, procurement exposure, or therapeutic-area composition. These controls are used to discipline or condition the market-return background, not to define MAH as a demand shock. With current approval-side data, the baseline moments are

$$m^{data} = \left( s_{post}^{E,data}, \bar{\lambda}_0^{O,obs,data} \right),$$

Table 3: Model Objects, Data Moments, and Claim Boundaries

| Model object                                        | Economic meaning                                                                                          | Feasible data moment                                                                                                                                           | Claim boundary                                                                                                                             |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| $R^{event}$ and $\bar{R}^E$                         | Route-independent gross return and entrusted-route realized-return kernel                                 | Demand-side controls: disease burden, patient population, reimbursement, procurement exposure, aggregate sales or therapeutic-composition controls             | Background/control for opportunity value; not evidence that MAH raises demand                                                              |
| $p_m^*(M)$                                          | Equilibrium price or effective payment for qualified contract-manufacturing services                      | Contract-manufacturing prices, CMO fee schedules, cost reports, producer capacity, or sensitivity values if unavailable                                        | Endogenous only in the CMO service market; not a product-market price index                                                                |
| Effective entrusted-route wedge containing $\tau^E$ | MAH-reducible policy, contractual, and regulatory friction embedded in the retained external route payoff | Holder–producer split among comparable original-drug records; post-MAH exposure-gradient in split shares                                                       | Enters a composite route-use wedge; does not separately identify legal cost, search cost, CMO price, or holder burden                      |
| $\zeta_i^E$                                         | Composite realization probability for the retained entrusted route                                        | Application-to-approval conversion, approval-to-launch conversion, launch or realization conversion; split shares discipline it only jointly with route choice | Distinguishes latent arrival from observed realization; not separately identified from $\bar{R}^E$ or route choice without additional data |
| $\mu_i^E$                                           | Residual holder-side monitoring, quality-control, technical-transfer, and coordination burden             | Not separately identified in approval-side data; handled by heterogeneity, normalization, or sensitivity values                                                | Prevents entrusted production from being treated as costless after MAH                                                                     |
| $\Gamma_i^M$                                        | Private expected value of a successful original-drug opportunity under regime $M$                         | Not directly observed; inferred through route-use and count moments after normalizations                                                                       | Private option value, not social welfare                                                                                                   |
| $x_i^*$                                             | Original-drug R&D intensity                                                                               | Usually unobserved in approval-side data; may be proxied by R&D spending, patents, or clinical-trial sponsorship in future modules                             | Not identified from approval records alone                                                                                                 |
| $\lambda_i^{lat,*}$ and $\lambda_i^{obs,*}$         | Latent opportunity arrival and observed realization                                                       | Original-drug applications, IND/CTA records, approvals, launches, or realized products, depending on data layer                                                | Approval counts are downstream realizations, not latent idea arrivals                                                                      |
| $n_{it}$                                            | Stock of retained original drugs or certificates                                                          | Holder-level stock of retained original-drug approvals or certificates                                                                                         | Captures retained assets; transferred products require separate interpretation                                                             |
| $N_E$ and $F_e$                                     | Research-oriented entry and entry-cost distribution                                                       | First appearance in original-drug applications, clinical-trial sponsorship, or holder records                                                                  | Future or sensitivity module unless firm-year entry data are merged                                                                        |

where  $s_{post}^{E,data}$  is the post-MAH holder–producer split share among comparable original-drug records and  $\bar{\lambda}_0^{O,obs,data}$  is a baseline realized original-drug count. A research-oriented entry moment  $N_E^{data}$  should be added only when a credible firm-year entry panel is available. The demand-side control vector  $d_t^{data}$  can be reported alongside these moments or used in sensitivity checks for  $R^{event}$  and  $\bar{R}^E$ .

**Step 2: Choose composite parameters.** The calibration should target composite objects rather than primitive costs:

$$\theta = (\omega_E, \alpha_O),$$

where  $\omega_E$  denotes a composite route-realization wedge that includes  $p_m^*$ ,  $\tau^E$ ,  $\mu^E$ ,  $\zeta^E$ , and the route-choice scale  $\sigma_r$  relative to internal production or another retained route, and  $\alpha_O$  denotes the composite innovation-arrival scale  $\beta a^2/\kappa$ . If a credible research-oriented entry panel is available, the parameter vector can be augmented with  $\mu_e$ , which indexes a parsimonious entry-cost distribution. Without such a panel,  $\mu_e$  is not part of the baseline approval-side calibration.

**Step 3: State normalizations.** Because approval-side data do not separately observe  $a_i$ ,  $\kappa$ ,  $\Gamma_i$ ,  $\zeta_i^E$ ,  $\bar{R}_i^E$ , project-level success probabilities, and route-choice scale, the calibration must normalize at least one object. Examples include  $\kappa = 1$ , a benchmark  $\Gamma^0(\theta) = 1$ , a chosen route-choice scale  $\sigma_r$ , or sensitivity values for  $p_m^*(M)$ . The calibrated parameters should be interpreted relative to these normalizations.

**Step 4: Match moments.** Let  $m^{model}(\theta)$  denote the moments generated by the model under the chosen normalizations. The calibration can minimize

$$Q(\theta) = [m^{data} - m^{model}(\theta)]' W [m^{data} - m^{model}(\theta)], \quad (63)$$

where  $W$  is a transparent weighting matrix. With few moments,  $W$  can be diagonal and sensitivity to alternative weights should be reported rather than hidden.

**Step 5: Report sensitivity and claim boundaries.** The baseline report should show how the calibrated composite route-realization wedge and innovation-arrival scale change under alternative original-drug definitions, sample restrictions, holder-producer name cleaning rules, route-use proxies, and sensitivity values for CMO prices or capacity. The exercise can support statements about consistency with the MAH commercialization-friction mechanism. It cannot separately estimate legal compliance costs, search costs, quality-agreement costs, contracting costs, contract-manufacturing prices, realization probabilities, post-production risk, or firm-specific project success probabilities.

### 6.3 Route-use moments under the logit model

The baseline theory already uses the logit route-choice model in equation (22). The route-use block is therefore not an extra smoothing trick; it is the empirical counterpart of the same route-choice structure. To isolate the logic, compare internal production and entrusted production among successful opportunities that remain with the innovating holder. In that conditional comparison,

$$G_t^E - G_t^I = \zeta_t^E(\bar{R}_t^E + v) - p_{m,t}^* - \tau_t^E - \mu^E - (R_t^{event} + v - C^I(k)). \quad (64)$$

Conditional on comparing retained internal production and retained entrusted production,

$$P_i(r = E \mid r \in \{I, E\}) = \frac{\exp(G_i^E/\sigma_r)}{\exp(G_i^E/\sigma_r) + \exp(G_i^I/\sigma_r)}. \quad (65)$$

Therefore,

$$\log\left(\frac{s_t^E}{1 - s_t^E}\right) = \frac{G_t^E - G_t^I}{\sigma_r}, \quad (66)$$

where  $s_t^E$  is a holder-producer split share among comparable retained original-drug records and  $\sigma_r$  is a route-choice scale. Without separate evidence on holder-side monitoring and quality-control costs, realization probabilities, route-specific returns, and CMO prices, route-use moments discipline a composite effective route-realization wedge rather than the primitive legal friction alone.

The baseline should not mechanically use a pre-post log-odds difference when the pre-MAH entrusted route is structurally unavailable. In that case  $s_{pre}^E = 0$  by institutional design, so  $\log(s_{pre}^E/(1 - s_{pre}^E))$  is not a comparable route-use moment. The baseline interpretation should treat the reform as route-set expansion plus a conditional friction decline. If the data permit, post-MAH exposure variation provides the cleaner calibration margin:

$$\log\left(\frac{s_{\ell,post}^E}{1 - s_{\ell,post}^E}\right) = \eta_0 + \eta_1 Exposure_\ell + u_\ell, \quad (67)$$

where  $\ell$  indexes an exposure cell, such as a region, dosage-form segment, firm type, or other implementation-intensity cell. The coefficient  $\eta_1$  disciplines relative differences in the effective entrusted-route wedge across exposure cells. A pseudo-count for a zero pre-MAH split share can be reported only as a sensitivity device, not as the baseline economic interpretation.

## 6.4 Count and entry blocks

Baseline original-drug counts discipline the composite R&D arrival scale and realization mapping. From equations (39) and (34),

$$\bar{\lambda}_0^{O,obs} = \mathbb{E} \left[ \frac{\beta a^2}{\kappa} \Gamma^0(\theta) \sum_{r \in \mathcal{R}_i(0)} P_i(r \mid 0, p_m^*(0)) \zeta_i^r(0, p_m^*(0)) \right]. \quad (68)$$

If the data are approval-side realized products, this moment should be described as a realized-product count rather than a direct observation of latent idea arrival.

Entrant counts discipline the entry-cost distribution only when the data contain a credible firm-year measure of research-oriented entry. From equation (51),

$$N_E(M) = \int F_e(A(\theta, M) \mid \theta) dG_N(\theta). \quad (69)$$

With only approval-side records, the entry block should be kept as a future module or sensitivity exercise.

## 6.5 Data requirements

The minimum product-level data are drug registration and approval records with product name, approval date, registration category, original-drug classification, holder, producer or manufacturer, dosage form, and indication or therapeutic class. These records are needed to construct original-drug counts and holder-producer split. Relevant official sources include NMPA and CDE registration and approval data, with careful cleaning of holder and producer names.

Demand-side data are useful as controls for  $R^{event}$  and  $\bar{R}^E$ . They may include disease burden, patient population, reimbursement status, NRDL status, procurement exposure, aggregate therapeutic-area sales, prescription volume, or therapeutic-area composition. These variables help keep the MAH commercialization-friction channel distinct from changes in market size or payment environment. They should not be interpreted as evidence that MAH itself raises demand.

The minimum firm-level data are legal-entity identifiers, founding date, location, ownership, business scope, observed first drug project, and proxies for research and manufacturing ability. Research ability  $a_i$  can be proxied by pre-reform original-drug applications, patents, clinical trial sponsorship, or R&D expenditure where available. Manufacturing ability  $k_i$  can be proxied by Drug Manufacturing Certificates, GMP or manufacturing-site information, dosage-form scope, production license records, or manufacturing assets. Firm registries such as the National Enterprise Credit Information Publicity System and commercial databases such as Tianyancha or Qichacha can help standardize firm identi-

ties.

The route-use outcome should identify whether the MAH or holder differs from the actual producer. The main proxy is

$$Split_p = \mathbf{1}\{Holder_p \neq Producer_p\}. \quad (70)$$

This proxy is not the policy treatment. It is an observed route-use proxy for realized products, and it should be validated against name standardization, group affiliation, and product-level regulatory records where possible. Data on quality agreements, technology-transfer requirements, audits, holder-manufacturer coordination, CMO prices, and application-to-approval status would be useful for separating parts of the composite route-realization value, but the baseline approval-side data should not be interpreted as doing that separation.

Useful additional data include CDE clinical trial or IND/CTA records, CNIPA patent data, centralized procurement data, hospital procurement or sales databases where accessible, and regional policy exposure measures. These data help separate the MAH commercialization channel from broader market-size or demand-side changes, but they are not required for the theoretical mechanism itself.

## 6.6 Future eight-moment extension

The baseline mechanism calibration should use only moment families supported by the available data. A richer future implementation can expand the exercise into eight empirical moments:

1. Original-drug holder-producer split share:

$$m_1 = \Pr(Split_p = 1 \mid O_p = 1).$$

This disciplines the composite route-realization wedge relative to the route-choice scale.

2. Realized original-drug count:

$$m_2 = \mathbb{E}[Y_{it}^O].$$

This disciplines the composite innovation-arrival scale and realization mapping.

3. Original-drug share among all realized products:

$$m_3 = \frac{Y_{it}^O}{Y_{it}^{all}}.$$

This disciplines the original-drug orientation of realized innovation.

4. External-route use among research-oriented or B-like firms:

$$m_4 = \Pr(Split_p = 1 \mid O_p = 1, B\text{-like firm}).$$

This tests whether the mechanism is strongest for firms most likely to need external commercialization.

5. Application-to-approval conversion for original drugs:

$$m_5 = \frac{\# \text{ Approved original drugs}}{\# \text{ Original-drug applications}}.$$

This moment should be used only after application and status data are available.

6. Research-oriented entry:

$$m_6 = \#\{\text{first-time original-drug applicants or sponsors}\}.$$

This disciplines the entry-cost distribution only when a credible firm-year entry panel is available.

7. Heterogeneity by pre-reform manufacturing capability:

$$m_7 = \Delta Y_{LowManu}^O - \Delta Y_{HighManu}^O$$

or

$$m_7 = \Delta Split_{LowManu}^O - \Delta Split_{HighManu}^O.$$

This checks whether the response is stronger among firms with weaker internal manufacturing capacity.

8. Producer-side entrusted-manufacturing response:

$$m_8 = \#\{\text{qualified entrusted producers}\}$$

or a related measure of producer-side participation. This is needed for the baseline CMO service-market closure when direct price data are unavailable.

These moments are not all available in the current approval-side data. The current baseline should use feasible approval-side moments and reserve conversion, entry, and detailed producer-side moments for future merged data modules or sensitivity checks.

## 7 Auxiliary Formal Details

The following sections collect auxiliary derivations and calibration details. They are formal support for the baseline model, not separate mechanisms added to the main text.

## 8 CES Background for the Commercialization Return

This appendix gives one possible background interpretation for the reduced-form commercialization return used in the main text. It is not part of the baseline market closure. Consider a representative payer or consumer allocating original-drug expenditure  $E_t$  across differentiated original-drug varieties  $\omega \in \Omega_t$  with CES preferences (Dixit and Stiglitz, 1977; Melitz, 2003):

$$X_t = \left[ \int_{\omega \in \Omega_t} (q_{\omega t} x_{\omega t})^{\frac{\sigma-1}{\sigma}} d\omega \right]^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1. \quad (71)$$

The associated quality-adjusted price index is

$$P_t = \left[ \int_{\omega \in \Omega_t} q_{\omega t}^{\sigma-1} p_{\omega t}^{1-\sigma} d\omega \right]^{\frac{1}{1-\sigma}}, \quad (72)$$

and demand for variety  $\omega$  is

$$x_{\omega t} = E_t P_t^{\sigma-1} q_{\omega t}^{\sigma-1} p_{\omega t}^{-\sigma}. \quad (73)$$

If the drug is priced with the CES markup, operating profit is

$$\varpi_{\omega t} = \frac{1}{\sigma} E_t P_t^{\sigma-1} q_{\omega t}^{\sigma-1} p_{\omega t}^{1-\sigma}. \quad (74)$$

The main text compresses this demand-side background into  $R^{event}$ . It does not solve for feedback from MAH-induced innovation to  $P_t$ , expenditure allocation, or product-market competition.

## 9 Post-Production Aggregate Risk

The main text uses  $\bar{R}_i^E$  as a compressed payoff object for the retained entrusted route. One interpretation is

$$\bar{R}_{ip}^E = \mathbb{E}_{\omega \sim F_{post}^E} [R_{ip}(\omega) \mid L_{ip} = 1, E]. \quad (75)$$

The shock  $\omega$  summarizes post-production uncertainty after the project has been implemented through the entrusted route. The model does not separately introduce sales, after-sales service, pharmacovigilance, regulatory penalty, or partner-performance states.

Those risks are intentionally compressed into  $\bar{R}^E$  so that the baseline remains a mechanism model rather than a full product-level dynamic state-space model.

## 10 Institutional Details and Mechanism Clarification

This appendix expands the institutional discussion behind the route-specific friction in the model. The objective is not to claim that MAH simply removes regulation. The point is more precise: MAH creates a legally recognized route under which marketing authorization can remain with the holder while production is performed by a qualified outside manufacturer, and it pairs that route with explicit holder-side responsibility, contractual requirements, quality-management obligations, and supervisory expectations.

### 10.1 Institutional evolution from pilot to legal consolidation

The institutional evolution matters because MAH is not a loose industry slogan. It is a progressively formalized legal arrangement. The reform path first weakened the tight practical bundling of research, authorization, and in-house production by allowing authorization-holding and production to be separated under regulated conditions ([General Office of the State Council of the People's Republic of China, 2016](#); [China Food and Drug Administration, 2016](#)). The later legal architecture then made the holder the entity associated with the drug registration certificate and located lifecycle responsibility on that holder ([National Medical Products Administration, 2019, 2022b](#)).

The key legal consolidation is that the holder may either manufacture by itself or entrust manufacturing to a qualified drug manufacturer. In the entrusted-production case, the holder and the manufacturer must sign both an entrustment agreement and a quality agreement, and they must perform the obligations specified in those agreements ([National Medical Products Administration, 2019](#)). Subsequent manufacturing supervision rules sharpen the operational meaning of this arrangement by requiring quality-assurance assessment, risk-management assessment, agreements, and supervision of performance ([National Medical Products Administration, 2022a, 2023](#)). In economic terms, the delegated-production route is not converted into a casual outsourcing margin. It is converted into a standardized, legally supported, and supervised route.

### 10.2 Holder responsibility after production is separated

For the economics of the paper, the important institutional point is that MAH separates production performance from ultimate holder responsibility. It does not transfer the core accountability for the drug away from the holder. Once a drug is authorized, the holder remains the focal entity for the drug across the life cycle.

At a minimum, four dimensions remain holder-side responsibilities:

1. **Lifecycle responsibility.** The holder remains the legally central entity from research and registration through production, distribution, post-marketing study, and pharmacovigilance.
2. **Quality-system responsibility.** The holder must establish and maintain a quality assurance system rather than rely passively on the contractor's internal systems.
3. **Oversight responsibility.** The holder must audit and supervise the quality-management systems and performance of entrusted manufacturers and other entrusted parties.
4. **Safety and release responsibility.** The holder remains the entity around which the law and supervisory rules organize quality, safety, and effectiveness responsibilities, even when physical production is performed elsewhere.

This institutional structure explains why the entrusted route is economically subtle. MAH did not create a zero-cost outsourcing technology. A research-originating holder cannot simply hand off production and walk away. In the model, this residual holder-side burden is captured by  $\mu_i^E$ . MAH can lower the policy and contractual friction  $\tau_i^E$  of using the retained external route, but it does not eliminate monitoring, quality-control, technical-transfer, or coordination burdens.

### 10.3 What the entrusted-route friction is not

The institutional record also clarifies what  $\tau^E$  and  $\mu^E$  are not.

**Not a demand shock.** The legal core of MAH concerns who may hold authorization, who may manufacture, how entrusted production is governed, and who bears quality responsibility. These are organizational and regulatory-allocation changes. They do not by themselves change reimbursement rules, final demand conditions, or therapeutic demand for a drug. Demand-side changes may coexist in the data through other policies, but they are not the distinctive institutional content of MAH.

This does not mean that market demand is absent from the model. Demand-side conditions enter through  $R^{event}$  and  $\bar{R}^E$  as background payoff shifters and calibration controls, while MAH operates through commercialization-route feasibility, realization probability, CMO service-market prices, and the relative attractiveness of the retained external route.

**Not a scientific-productivity shock.** Nothing in the MAH architecture changes the scientific technology by which an idea is discovered. The reform may raise expected payoff to research because more ideas become commercializable or because a route becomes more feasible, but that is an expected-return channel, not a direct productivity shift in the scientific production function.

**Not a generic fixed-cost reduction for all firms.** MAH does not simply relax every firm's operating burden. The post-law architecture may impose more explicit documentation, auditing, and quality-system obligations on the holder. What changes is that these obligations are attached to a legally recognized and standardized external route, so that the MAH-reducible component  $\tau_i^E$  can fall relative to the pre-reform environment. The residual component  $\mu_i^E$  remains in the entrusted-route payoff. In other words, MAH can preserve or sharpen holder-side responsibility while still lowering the policy and contractual cost of external implementation by making the arrangement legally supported, contractually structured, and regulatorily intelligible.

This is why the model separates  $p_m^*(M)$ ,  $\tau_i^E$ ,  $\mu_i^E$ , and  $\zeta_i^E$ . The direct service price  $p_m^*(M)$  is distinct from the MAH-reducible policy friction  $\tau_i^E$ , both are distinct from residual holder-side monitoring and quality-control burden  $\mu_i^E$ , and  $\zeta_i^E$  captures route realization rather than primitive legal cost. The comparative statics in the model should therefore be read as statements about the private option value of a legally recognized entrusted-production route, not as statements about demand, science, or a generic deregulation shock.

## 11 Baseline Contract-Manufacturing Service Market

This appendix expands the CMO service-market closure used in the baseline model. It is the only market-clearing condition in the paper.

The demand for entrusted manufacturing services is

$$D_m(p_m; M) = \int a_i x_i^*(M, p_m) P_i(r = E \mid M, p_m) \zeta_i^E(M, p_m) dH_i, \quad (76)$$

where  $a_i x_i^*(M, p_m)$  is latent opportunity arrival,  $P_i(r = E \mid M, p_m)$  is the logit route probability, and  $\zeta_i^E(M, p_m)$  maps latent entrusted-route use into expected realized entrusted-production demand.

The supply of qualified contract-manufacturing services is

$$S_m(p_m) = \int s_j(p_m; z_j^C) dH_C(j), \quad (77)$$

where  $z_j^C$  indexes producer-side manufacturing capability.

The service-market-clearing condition is

$$D_m(p_m; M) = S_m(p_m). \quad (78)$$

Equation (78) gives  $p_m^*(M)$ . A full general-equilibrium model would additionally require product-market clearing, R&D input-market clearing, free entry, exit, and an

invariant firm distribution. The present paper deliberately does not take those steps.

## 12 GHM-Style Outside Option and Hold-Up Extension

The baseline treats the non-retained transfer value  $S_i$  as exogenous. This is conservative because MAH may improve the innovator’s bargaining position even when the innovator ultimately transfers, sells, or out-licenses the project. In the property-rights logic of [Grossman and Hart \(1986\)](#) and [Hart and Moore \(1990\)](#), control over an asset can strengthen outside options and reduce hold-up risk.

A simple reduced-form extension is

$$S_i(M) = \bar{S}_i + \rho_T \max\{G_i^E(M), 0\}, \quad \rho_T \in [0, 1]. \quad (79)$$

This does not introduce a full Nash bargaining model. It only records the idea that the credible ability to retain authorization and use entrusted production can raise the innovator’s threat point in transfer negotiations. If  $\rho_T > 0$ , the baseline model understates the total MAH effect because MAH raises not only the retained entrusted route value but also the non-retained transfer outside option.

## 13 Overview of Calibration Strategy

This appendix documents the mechanism-calibration blueprint behind the calibration section. The objective is not to turn the theory section into a full dynamic structural estimation exercise, nor to claim a complete causal empirical design. The objective is narrower: to use a small set of empirical moments that are close to the mechanism to discipline a small set of composite model objects.

The current approval-side calibration targets two baseline objects and one future object:

- (i) the effective route-realization wedge, which combines  $p_m^*(M)$ ,  $\tau^E$ ,  $\mu^E$ ,  $\zeta^E$ ,  $\bar{R}^E$ , and the route-choice scale;
- (ii) the composite scale that converts R&D intensity into latent original-drug opportunities and then into observed realizations;
- (iii) the distribution of research-oriented entry costs, kept as a future or sensitivity module when the current data do not contain a firm-entry panel.

Demand-side controls help proxy or condition  $R^{event}$  and  $\bar{R}^E$ . They are not a fourth

MAH mechanism and should not be interpreted as evidence that MAH increases demand.

Therefore, the calibration section should be read as a **mechanism calibration blueprint**, not as full structural identification. It asks whether observed route-use patterns and realized original-drug outcomes are consistent with the mechanism that MAH expanded the feasible commercialization route set, changed route-realization probability, lowered the MAH-reducible policy and contractual friction of the external commercialization route, and induced a CMO price response. It does not claim to separately identify legal compliance costs, search costs, residual monitoring burdens, quality-management costs, contract-manufacturing prices,  $\zeta^E$ ,  $\bar{R}^E$ , latent idea arrivals, or firm-specific project success probabilities.

## 14 Definition of Moments

A moment is a statistic that can be computed both in the data and in the model. For example, in the data one may compute

$$s_{post}^{E,data} = \frac{\#\{\text{post-reform original-drug records with different holder and producer}\}}{\#\{\text{comparable post-reform original-drug records}\}}.$$

Given parameters, the model can generate the corresponding object:

$$s_{post}^{E,model}(\theta).$$

The calibration logic is to choose parameters  $\theta$  such that

$$m^{model}(\theta) \approx m^{data}.$$

The moment  $m$  can be a route-use share, an original-drug count, an entrant count, an original-drug share, or a holder–producer split. A moment is not a new structural variable. It is the empirical bridge between the model and the data.

## 15 Calibration Blocks

The three calibration blocks in the calibration section are summarized in Table 4.

Table 4: Three Calibration Blocks

| Calibration block    | Data moment                                                                                            | Model moment                                                                                                                          | Disciplined object                                                   |
|----------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Route-use data       | Post-MAH holder–producer split share; exposure-gradient in split shares; pseudo-count sensitivity only | $\log \frac{s_{\ell, post}^E}{1-s_{\ell, post}^E} = \eta_0 + \eta_1 Exposure_{\ell} + u_{\ell}$                                       | Composite route-realization wedge or exposure-gradient in that wedge |
| Original-drug counts | Baseline original-drug approvals, realized products, or applications                                   | $\bar{\lambda}_0^{O, obs} = \mathbb{E}[\frac{\beta a^2}{\kappa} \Gamma^0(\theta) \sum_r P_i(r   0, p_m^*(0)) \zeta_i^r(0, p_m^*(0))]$ | Composite innovation-arrival scale plus realization mapping          |
| Entrant counts       | Research-oriented entrant counts or proxies, only with a credible firm-year panel                      | $N_E(M) = \int F_e(A(\theta, M)   \theta) dG_N(\theta)$                                                                               | Future module: entry-cost distribution $F_e$                         |

These blocks correspond to the same mechanism chain:

$$\begin{aligned}
& \tau^E \downarrow, \quad \zeta^E \text{ may improve, } p_m^* \text{ adjusts,} \\
& \Rightarrow \Gamma \text{ rises only when net entrusted-route value rises,} \\
& \Rightarrow x^* \text{ and } \lambda^{lat,*} \text{ respond through } \Gamma, \\
& \Rightarrow \lambda^{obs,*} \text{ may rise through route use and realization,} \\
& \Rightarrow \Lambda^{lat,agg} \text{ may rise for fixed incumbents,} \\
& \Rightarrow N_E \text{ remains an optional entry-module response.}
\end{aligned}$$

Here  $\tau^E$  is the MAH-reducible policy and contractual friction of the external commercialization route,  $\mu^E$  is the residual holder-side monitoring and quality-control burden,  $\zeta^E$  is the composite realization probability,  $\Gamma$  is the logit inclusive value of a successful original-drug opportunity,  $x^*$  is optimal original-drug R&D intensity,  $\lambda^{lat,*}$  is the latent arrival rate of original-drug opportunities,  $\lambda^{obs,*}$  is the realized-output counterpart,  $\Lambda^{lat,agg}$  is aggregate expected latent opportunities for a fixed incumbent distribution, and  $N_E$  is research-oriented entry only in the optional entry module.

## 16 Route-Use Moment and Entrusted-Production Wedge

### 16.1 The Data Moment

The central data object for route use is the holder–producer split. For product record  $p$ , define

$$Split_p = \mathbb{I}\{Holder_p \neq Producer_p\}.$$

Restricting attention to original-drug records, define

$$s_t^{E,data} = \frac{\sum_{p \in t} Split_p O_p}{\sum_{p \in t} O_p},$$

where  $O_p = 1$  if product  $p$  is classified as an original drug or satisfies the paper's baseline original-drug proxy. This share is not the policy treatment. It is an observed outcome that proxies the use of an external commercialization route while retaining authorization.

## 16.2 Why Route Share Maps to the Entrusted-Route Wedge

After the reform, when the entrusted route is feasible, a firm compares commercialization routes after a successful original-drug opportunity is realized. To isolate the logic, consider only internal production and entrusted production among retained products:

$$G_t^I = R_t^{event} - C^I(k) + v,$$

$$G_t^E = \zeta_t^E (\bar{R}_t^E + v) - p_{m,t}^* - \tau_t^E - \mu^E.$$

Subtracting internal production from entrusted production gives

$$G_t^E - G_t^I = \zeta_t^E (\bar{R}_t^E + v) - p_{m,t}^* - \tau_t^E - \mu^E - (R_t^{event} + v - C^I(k)).$$

The older special case in which  $R_t$  and  $v$  cancel is obtained only when  $\zeta_t^E = 1$  and  $\bar{R}_t^E = R_t^{event}$ . In the baseline model, route share disciplines a composite route-realization value that includes the realization probability, route-specific realized return, CMO service price, MAH-reducible route friction, and residual holder-side burden.

Using the same logit route-choice model as in the main text,

$$s_t^E = \frac{\exp(G_t^E / \sigma_r)}{\exp(G_t^E / \sigma_r) + \exp(G_t^I / \sigma_r)}.$$

Then the odds ratio is

$$\frac{s_t^E}{1 - s_t^E} = \exp\left(\frac{G_t^E - G_t^I}{\sigma_r}\right),$$

and taking logs yields

$$\log\left(\frac{s_t^E}{1 - s_t^E}\right) = \frac{G_t^E - G_t^I}{\sigma_r}$$

This equation does not say that the route-use share equals  $\tau^E$ ,  $\mu^E$ , or  $\zeta^E$ . It says that the route-use share, through a retained-product route-choice equation, disciplines the relative attractiveness of the entrusted route. Without additional data on internal production costs, direct contract-manufacturing costs, quality agreements, audit costs, contract terms, firm-level regulatory execution, and application-to-approval conversion,

the calibration disciplines only a composite route-realization wedge relative to the route-choice scale.

### 16.3 When a Pre-Post Log-Odds Moment Is Valid

A pre-post log-odds moment is valid only if the pre-reform data contain a legally and economically comparable entrusted-retention route. Under that condition, and if internal-production costs, CMO prices, residual holder burdens, realization values, and sample composition are stable across the comparison or handled through comparable samples and controls, one can write

$$\frac{\Delta\omega_E}{\sigma_r} = \log\left(\frac{s_{post}^E}{1 - s_{post}^E}\right) - \log\left(\frac{s_{pre}^E}{1 - s_{pre}^E}\right),$$

where

$$\Delta\omega_E = (G_{post}^E - G_{post}^I) - (G_{pre}^E - G_{pre}^I).$$

The object  $\Delta\omega_E$  is a change in the relative entrusted-route payoff wedge, not the primitive change in  $\tau^E$  alone. It equals  $\tau_{pre}^E - \tau_{post}^E$  only under a restrictive normalization that holds CMO prices, residual holder burden, internal production cost, realization probability, route-specific realized return, and retained continuation value fixed. This is not the baseline interpretation when the MAH-style entrusted route was institutionally unavailable before the reform.

### 16.4 What If Pre-Reform Route-Use Data Are Unavailable?

This is the central practical issue. If the MAH-style entrusted route was institutionally unavailable before the reform, then there is no comparable pre-reform entrusted-route share. Writing

$$s_{pre}^E = 0$$

as though it were an observed route-use rate is misleading, because

$$\log\left(\frac{s_{pre}^E}{1 - s_{pre}^E}\right) = \log(0)$$

is undefined. The formula cannot be used mechanically. There are three disciplined responses.

**Response 1: Treat the pre-reform regime as a route-set-expansion boundary case.** Before MAH,

$$\mathcal{R}_i(0) = \{A, T\} \cup \{I : h_i^I = 1\}.$$

After MAH,

$$\mathcal{R}_i(1) = \{A, T\} \cup \{I : h_i^I = 1\} \cup \{E : q_i^E = 1\}.$$

Thus there is no comparable pre-reform entrusted-route share. The baseline interpretation should emphasize that MAH first expands the feasible route set and then, conditional on route feasibility, lowers the MAH-reducible policy and contractual friction of the route.

**Response 2: Use post-MAH exposure or implementation-intensity variation.**

If the route is unavailable before MAH, the preferred empirical calibration margin is post-MAH variation across regions, dosage forms, therapeutic classes, or firm types. For example, regions or dosage-form segments with greater pre-reform qualified manufacturing capacity may experience a larger effective decline in external commercialization friction. A transparent reduced-form calibration moment is

$$\log \left( \frac{s_{\ell, post}^E}{1 - s_{\ell, post}^E} \right) = \eta_0 + \eta_1 Exposure_{\ell} + u_{\ell},$$

where  $\ell$  indexes an exposure cell, such as a region, dosage-form segment, firm type, or other implementation-intensity cell. This disciplines relative differences in the effective entrusted-route wedge without imposing a nonexistent pre-reform route-use rate.

**Response 3: Use a small pseudo-count only as a sensitivity device.** For example, use

$$\tilde{s}_{pre}^E = \frac{0.5}{N_{pre} + 1}$$

instead of zero. This makes the log-odds expression computable, but the pseudo-share should not be interpreted as a real pre-reform route-use rate. It should be reported only as a sensitivity check, not as the baseline calibration moment.

## 16.5 What This Block Can and Cannot Identify

This block can support the following statement: if the holder–producer split among original-drug records rises after MAH, or if higher-exposure segments show higher post-MAH split shares, and this pattern cannot be explained by changes in product composition, therapeutic class, region, or firm capability composition, the pattern is consistent with a higher relative route-realization value of the entrusted route.

It cannot support the following stronger statement: the data separately identify legal compliance costs, search costs, quality-agreement costs, contracting costs, contract-manufacturing prices, realization probabilities, or post-production risk-adjusted returns. These components enter the route-choice payoff as a composite wedge. Without additional data on those objects, they cannot be separately recovered.

## 17 Original-Drug Counts and R&D Arrival Scale

### 17.1 The Model Equation

The firm chooses original-drug R&D intensity  $x_i$ . The latent arrival rate of original-drug opportunities is

$$\lambda_i(x_i) = a_i x_i.$$

Given the expected commercialization value  $\Gamma_i^M$  of a successful opportunity, the first-order condition gives

$$x_i^*(M) = \frac{\beta a_i}{\kappa} \Gamma_i^M.$$

Therefore, the optimal latent arrival rate is

$$\lambda_i^{lat,*}(M) = a_i x_i^*(M) = \frac{\beta a_i^2}{\kappa} \Gamma_i^M.$$

Observed realized output further multiplies latent arrival by route-choice and realization probabilities as in equation (34). Taking expectations in the pre-reform or baseline regime gives

$$\bar{\lambda}_0^{O,obs} = \mathbb{E} \left[ \frac{\beta a_i^2}{\kappa} \Gamma^0(\theta) \sum_{r \in \mathcal{R}_i(0)} P_i(r \mid 0, p_m^*(0)) \zeta_i^r(0, p_m^*(0)) \right].$$

### 17.2 The Data Moment

The most direct moment is the baseline number of original-drug outcomes:

$$\bar{\lambda}_0^{O,data} = \frac{\#\{\text{pre-reform original-drug records}\}}{\#\{\text{pre-reform years}\}}.$$

At the firm-year level, define

$$Y_{it}^{O,base} = \sum_{p \in (i,t)} O_p^{base}.$$

If the current dataset is an approval-side realized-product file, this object should be interpreted as realized original-drug products, not latent idea arrival. If future data include CDE applications, IND/CTA records, or clinical-trial registrations, those objects can be used to construct moments closer to upstream innovation-opportunity arrival.

### 17.3 Why This Moment Only Disciplines a Composite Parameter

From

$$\lambda_i^{obs,*} = \frac{\beta a_i^2}{\kappa} \Gamma_i \sum_r P_i(r) \zeta_i^r,$$

original-drug counts depend on at least four primitive or composite object groups:

- (i)  $a_i$ : research ability;
- (ii)  $\kappa$ : the marginal cost scale of R&D effort;
- (iii)  $\Gamma_i$ : the commercialization value of a successful opportunity;
- (iv) route-use and realization probabilities.

If the data only reveal realized original-drug counts, the model cannot separately identify  $a_i$ ,  $\kappa$ ,  $\Gamma_i$ , and  $\zeta_i^r$ . High original-drug counts may reflect strong scientific capability, low R&D effort costs, high commercialization value, high route-use probability, or high realization probability. Without project-level R&D costs, project success rates, project values, application-to-approval status, or rich firm-capability data, these objects cannot be separately recovered.

The disciplined approach is to normalize one object. For example,

$$\kappa = 1,$$

or for a benchmark firm,

$$\Gamma^0(\theta) = 1.$$

Then

$$\frac{\beta a^2}{\kappa}$$

is interpreted as a composite innovation-arrival scale, while realized-product counts also require assumptions on route-use and realization probabilities.

## 17.4 What If Pre-Reform Original-Drug Counts Are Sparse?

Sparse pre-reform approvals do not invalidate the mechanism. Drug development cycles are long, and final approvals are downstream outcomes. Pre-reform approval counts can therefore be noisy low-frequency moments. Disciplined responses include:

1. using multi-year averages rather than year-by-year moments;
2. using IND/CTA records, clinical-trial registrations, CDE acceptance numbers, or original-drug applications as upstream proxies;
3. separating realized approvals from upstream applications as distinct moments;
4. reporting robustness under alternative original-drug definitions, such as broad new-drug definitions, strict class-I innovative drugs, or inclusion/exclusion of improved new drugs.

The writing boundary is simple: approval-side data calibrate realized original-drug product counts; application-side data are closer to innovation-opportunity arrival.

## 18 Entry Counts and Entry-Cost Distribution

### 18.1 The Model Equation

A potential entrant enters if post-entry value exceeds entry cost:

$$A(\theta, M) \geq f_e.$$

The post-entry value is

$$A(\theta, M) = \frac{\beta^2 a^2}{2\kappa(1 - \beta)} [\Gamma^M(\theta)]^2.$$

If  $f_e$  has conditional distribution function  $F_e(\cdot | \theta)$ , the entry probability is

$$\Pr(f_e \leq A(\theta, M) | \theta) = F_e(A(\theta, M) | \theta).$$

Aggregating over potential entrants gives

$$N_E(M) = \int F_e(A(\theta, M) | \theta) dG_N(\theta).$$

For a fixed potential entrant, the strict entry-response region is

$$f_i^e \in [A(\theta_i, 0), A(\theta_i, 1)).$$

This threshold statement mirrors the firm-level route-choice result: MAH changes entry only for potential entrants whose entry costs are between the pre-MAH and post-MAH values. In the logit baseline,  $\Gamma^M$  is the inclusive value rather than a deterministic maximum.

### 18.2 The Data Moment

The ideal data moment is the number of research-oriented entrants, not the number of all newly registered pharmaceutical firms. Examples include:

- firms that appear for the first time in original-drug applications;
- firms that sponsor clinical trials for the first time;
- firms that first appear as MAH/holder for original-drug records;
- firms without traditional production capacity that enter original-drug R&D or com-

mercialization records.

Define

$$Entry_{it}^R = 1$$

if firm  $i$  first enters observable original-drug R&D or original-drug realization activity in year  $t$ . Then the data moment is

$$N_{E,t}^{data} = \sum_i Entry_{it}^R.$$

### 18.3 Why This Step Should Be Sensitivity or a Future Module

A current approval-side realized-product file is typically not a firm-entry panel. Observing that a firm first appears in an approval record is not equivalent to observing true firm entry or research-oriented entry. There are three reasons:

1. the firm may have been founded long before its first approval;
2. the firm may have upstream R&D activity that has not yet reached final approval;
3. name standardization, acquisitions, subsidiaries, and holder-entity changes can affect the first observed year.

Therefore, entrant-count calibration should not be a baseline result in the current data environment. It should be kept as a future data module or as a sensitivity appendix. A serious calibration of  $F_e$  requires a merged firm-year entry panel, business registration data, CDE application data, and firm-capability data.

### 18.4 How to Calibrate Entry Costs

One can assume a simple one-parameter distribution. For example,

$$f_e \sim \text{Exponential}(\mu_e),$$

so that

$$F_e(A) = 1 - \exp(-A/\mu_e).$$

With a credible firm-year entry panel, the pre-MAH entry rate can be used to match  $\mu_e$ , and the post-MAH change in  $\Gamma^M$  can then be used to predict the post-MAH entry response. Without such a panel,  $\mu_e$  is only a sensitivity parameter. A lognormal distribution is also possible, but if the data are thin, it is better not to introduce too many shape parameters.

The interpretation should be modest: this step does not estimate each potential entrant's true entry cost. It checks whether the entry margin moves in the direction

predicted by a lower external commercialization wedge.

## 19 Handling Missing Pre-Reform Data

Missing or weak pre-reform data are not a minor nuisance. They are central to the calibration design. The empirical architecture should therefore be modular.

### 19.1 Module A: Current Approval-Side Realized-Product Moments

The most credible baseline moments are:

1. realized original-drug product counts;
2. original-drug share among realized products;
3. holder–producer split;
4. split-based route-use proxy among original-drug records.

These objects can be constructed from the current approval-side matching file, but they are realized-product objects rather than full R&D-process objects.

### 19.2 Module B: Future Application and Status Panel

Once CDE applications, IND/CTA records, clinical trials, withdrawn projects, pending projects, rejected projects, and approved projects are merged, one can construct:

1. application-to-approval conversion;
2. project realization probabilities;
3. upstream original-drug opportunity arrival;
4. delay and conversion moments.

These moments are closer to  $\lambda_i$  and commercialization success.

### 19.3 Module C: Future Firm-Capability and Entry Panel

Once business-registration records, production licenses, GMP or dosage-form scope, R&D expenditure, patents, and clinical-trial sponsorship are merged, one can construct:

1. pre-reform research capability;
2. pre-reform manufacturing capability;

3. research-oriented entry;
4. capability-bin-specific route-use and realization moments.

These moments are needed to calibrate the firm-characteristic distribution,  $F_e$ , and heterogeneous responses more credibly.

## 20 Technical Appendix Conclusion

The full model is complete only when the three layers are kept distinct. The main paper states the mechanism and core implications. This technical appendix records the formal derivations and calibration boundaries. The model notes preserve design decisions, dependency checks, and future modifications before they are promoted into the appendix or the main text.

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